Thermal effects and energy budgets

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a temperature change only after stating how much energy enters the material.

SUCCESS CRITERIA - I can...

1. Apply an explicit absorbed-energy fraction.
2. Use mass and specific heat in a temperature estimate.
3. State the no-phase-change and no-loss assumptions.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “absorbed” and “specific heat” to explain your model or result. Sentence starters are optional.

absorbed: taken into the specified object or system

specific heat: energy per mass needed for a one-degree temperature change

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

A refrigerator uses 15 J electrical work while moving heat from its colder interior to a warmer room.
Explain why this is not spontaneous cold-to-hot heat flow.

Word help: spontaneous = able to occur without the specified external work input

Sentence starter: My claim is ... . The diagram/data show ..., so ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q_{absorbed} = mc\,\Delta T$; $\Delta T = Q_{absorbed}/(mc)$.
First identify the absorbed energy. Use only when there is no phase change and c is approximately constant.

Worked reasoning: A surface receives 500 J. A model accounts for 200 J reflected, 250 J absorbed as heating and 50 J transmitted. The full account sums to 500 J.

Your turn: Draw proportional energy arrows and identify which part can directly raise this surface's temperature.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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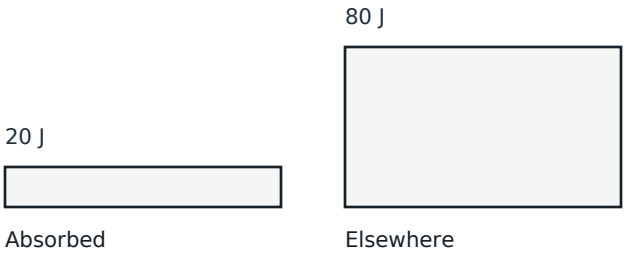
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A model releases 100 J. Only 20% enters a sample.
Draw the energy partition and calculate energy absorbed.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

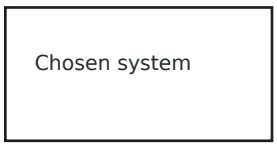
Marzano check - circle: 1 2 3

04 Apply the physics

In a heating simulation, a 1 kg sample with c = 10 J/(kg·°C) absorbs 20 J. Assume no phase change or loss.
Label absorbed energy on the model and calculate ΔT.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3

05 Explain your reasoning

A temperature prediction exceeds a material's melting point, but the calculation used only mc ΔT. What must be added or reconsidered?

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A device receives 60 J and delivers 42 J useful mechanical energy.
Find other energy and the useful fraction.
Show the energy changes and transfers; label units.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

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Continue down each column.

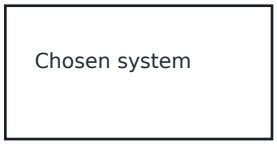
Sentence starters are optional. You may plan in any language.

07 Exit check

If absorbed energy doubles for the same mass and material with no phase change, compare temperature rises.
Label energy entering, leaving or stored on the system model.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

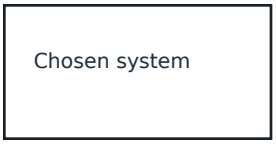
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Two surfaces absorb 250 J but have different masses of the same material.
Compare their temperature rises if one mass is twice the other.
Label energy entering, leaving or stored on the system model.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: 1 2 3